

Luke Kulm

lbk73@cornell.edu | San Francisco, CA | lukekulm.github.io

EDUCATION

Cornell University, College of Engineering, *B.S. in Computer Science*, GPA: 3.73

May 2025

- Coursework: Systems Programming, Analysis of Algorithms, Machine Learning, Computer Vision, Robot Learning, Large Language Models

EXPERIENCE

Bucket Robotics, *Machine Learning Engineer* | San Francisco, CA

Oct. 2025 – Present

- Reduced computer-vision inference latency on legacy Intel industrial hardware from ~50 seconds to ~2 seconds (20x+) by profiling ONNX Runtime and browser bottlenecks, evaluating quantized model variants with latency and accuracy/parity tests, and building custom WebAssembly/WebGPU execution paths
- Built an ML deployment platform that converts CAD models into synthetic training data and orchestrates training, evaluation, and rollout for hundreds of models across dozens of manufacturing deployments
- Translated research in domain alignment, augmentation, and diffusion-based generation into production pipelines, improving real-world accuracy by ~20% and enabling deployments with only tens of real images
- Built and maintained production services spanning PostgreSQL/Ruby on Rails data models, model deployment tooling, and customer-facing inspection interfaces in a shared monorepo
- Owned deployment and debugging across model, backend, and customer-facing systems for ~7 customers; converted hardware and operational constraints into reliable engineering plans

Yrikka, *Machine Learning Engineering Intern* | New York, NY

May 2024 – Sept. 2024

- Built an agentic evaluation pipeline to benchmark the accuracy and robustness of multimodal vision-language models for healthcare applications
- Evaluated vision encoders under FGM, PGD, and adversarial patch attacks and investigated linear probes on LLM hidden states for more interpretable, deterministic evaluation

Cornell University, *Course Creator and Teaching Assistant*, *CS 4782* | Ithaca, NY

Aug. 2023 – May 2025

- Helped build Cornell's first undergraduate deep learning course and taught and reviewed student PyTorch implementations as the course scaled
- Developed programming assignments and evaluation tests covering vision transformers, diffusion models, and reinforcement learning

Emprise Robotics Lab, *Undergraduate Researcher* | Ithaca, NY

May 2023 – May 2024

- Developed multimodal perception models over video, audio, and force data using Python, PyTorch, ROS, and Linux for a human-in-the-loop robotic peeling system
- Wrote C code for an Arduino to ingest and process real-time sensor data and integrated the resulting signals into the robot pipeline

PUBLICATIONS AND PROJECTS

MORPHeus: Multimodal One-Armed Robot-Assisted Peeling | **ICRA 2024**

2024

- Co-authored a peer-reviewed robotics paper and built the ROS/PyTorch perception stack integrating video, audio, and force sensing
- Integrated multimodal predictions with CAD geometry and natural-language instructions to support autonomous peeling actions

Adversarial Robustness of I-JEPA

2024

- Benchmarked Meta's I-JEPA representation model under FGM, PGD, and adversarial patch attacks across perturbation strengths
- Analyzed representation shifts using cosine distance and published a technical article explaining the methodology and results

Robot Learning from Human Preferences

2024

- Implemented a modular deep-RLHF pipeline spanning preference collection, reward modeling, and PPO policy training
- Trained robotic manipulation policies from learned human preferences without a hand-designed reward function

SKILLS

ML Systems

PyTorch, ONNX, model deployment, inference optimization, computer vision, synthetic data, model evaluation, agentic workflows

Languages

Python, C/C++, JavaScript, SQL, Ruby

Infrastructure

WebGPU, WebAssembly, Linux, Docker, AWS, PostgreSQL, Ruby on Rails, Git